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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Multi Disciplinary Elective (MDE)

MG1MDEBCA100-CYBER LAWS AND SECURITY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

ANSWER KEY

Part A

Very Short Answer Questions

Answer **All** Questions

Each Question carries **2** marks

1. Explain the impact of IT Act amendments on digital signatures.? **K/CO1**
   The amendments to India's IT Act have significantly impacted digital signatures by providing them with the **same legal validity as handwritten signatures** for most transactions, ensuring contracts signed electronically are legally enforceable.
2. List out the rights each individual have over their data? **K/CO1**
   - Right to access
   - Right to rectification
   - Right to erasure
   - Right to restrict processing
   - Right to data portability
3. Identify the purpose of Cyber laws. **U/CO1**
   - Cyber crime prevention and regulation
   - Data protection and privacy
   - Intellectual property rights in cyberspace

- Regulation of social media and content platforms
- E-Commerce

4. How does checking for a padlock icon help in avoiding phishing attacks? **A/CO2**

Checking for a padlock icon helps in avoiding phishing attacks by indicating that the connection to the website is encrypted and secure, specifically by using the HTTPS protocol.

To avoid phishing attacks,online user needs to keep the following points

Encryption, Authentication (Domain Validation), Always check the URL etc.

5. How can you implement measures to safeguard your personal information while using online banking? **A/CO2**

To safeguard personal information during online banking, create strong, unique passwords, enable multi-factor authentication, and avoid public Wi-Fi.

- Use strong, unique passwords
- Change passwords regularly
- Enable multi-factor authentication (MFA)
- Log out the account completely:

6. Describe the term "Internet Security." **K/CO2**

Internet security is the practice of protecting internet-connected systems and data from cyber threats, such as unauthorized access, data theft, and malicious attacks.

7. How does cyber forensics assist in legal proceedings? **U/CO3**

- Data acquisition and Preservation
- Attack Identification
- Tracing Cyber criminals
- Legal Support
- Recovering and Decrypting data

8. What do you mean by Jurisdiction? **K/CO1**

Jurisdiction is the authority of a court, government, or other legal body to interpret and apply the law and dispense justice within a specific geographical area or over a particular subject matter.

9. Examine the main goal of digital forensics **AN/CO3**

- Prevent hackers
- Prevent malware
- Information recovery
- Cyber crime Investigation
- Identifying vulnerabilities

10. Explain the importance of cyber forensics in solving cybercrimes **U/CO3**

Cyber forensics is crucial for solving cybercrimes because it provides a systematic process for collecting, analyzing, and preserving digital evidence. This evidence is used to identify perpetrators, understand how attacks happened, recover lost data, and support legal action against those who commit crimes like hacking, identity theft, and fraud.

Part B

Short Answer Questions

Answer any **6** Questions

Each Question carries **5** marks

11. Discuss the importance of social media policies in preventing online harassment. **U/CO1**

Social media policies are crucial for preventing online harassment by establishing clear guidelines, enabling user reporting, and leveraging technology for moderation. They hold platforms accountable for user safety by setting rules against behavior like cyberbullying and hate speech, while also providing users with tools to block harassers and report harmful content. The combination of clear policies, user-driven reporting, and platform-side moderation creates a more accountable and safer online environment.

12. List the cybercrimes addressed under Sections 66, 66B, 66C, 66D, and 66F of the IT Act, along with their penalties? **K/CO1**

Section 66	Offences associated with computers or any act outlined in Section 43	Imprisonment of 3 years or a fine that extends to Rs. 5 lakhs or both
Section 66B	Dishonestly receiving a stolen computer source or device	Imprisonment for 3 years or a fine of Rs. 1 lakh or both
Section 66C	Identity theft	Imprisonment of 3 years or a fine of Rs. 1 lakh or both
Section 66D	Cheating by personation	Either imprisonment for 3 years or a fine of Rs. 1 lakh or both

Section 66E	Invading privacy	Either imprisonment up to 3 years or a fine of Rs. 2 lakhs or both
Section 66F	Cyber terrorism	Life imprisonment

13. Explain the two methods to prevent or respond to cyberbullying? **K/CO1**

The two primary methods for addressing cyberbullying are **prevention** through proactive measures and **response/intervention** after an incident has occurred

1. **Prevention Methods** Prevention focuses on building skills and creating a safer online environment to stop cyberbullying before it starts.

- **Education and Awareness** Teaching about responsible online behavior, digital citizenship, and the consequences of cyberbullying helps individuals understand what it is and how to avoid being a part of it, either as a victim or a perpetrator.

- **Privacy and Security** Using strong passwords and adjusting privacy settings on social media and other platforms limits access to personal information, making it harder for potential bullies to target someone.

- **Open Communication** Maintaining open and frequent communication with children about their online activities and feelings ensures they feel comfortable coming to a trusted adult if an issue arises.

2. **Response methods** These are actions taken when cyberbullying is actively happening or has already occurred to stop the behavior and support the victim.

- **Do Not Respond or Retaliate** Engaging with a cyberbully often escalates the situation. The best approach is to ignore the messages and avoid an emotional reaction.
- **Document and Save Evidence** Take screenshots and keep a record of all harassing messages, posts, or images. This "digital footprint" is crucial evidence when reporting the incident to authorities, schools, or the platform administrators.
- **Block the Bully** Use built-in tools on social media, messaging apps, and phones to block the person from contacting you further.

14. List two benefits of using MFA?

MFA provides extra layers of security, protecting against unauthorized access even if passwords are compromised, while also helping organizations meet industry regulations that require strong data protection.Two benefits of using Multi-Factor Authentication (MFA) are

1. Enhanced security

2. Regulatory compliance.

15. Describe various types of information social media platforms collect about you? **U/CO2**

Social media platforms collect a wide variety of data, including **profile information** (name, age, location), **content you create** (posts, photos, messages), **user activity** (likes, shares, comments), and **behavioral data** (time spent, clicks, browsing habits). They also collect technical data about your **devices** (IP address, device ID) and **location**, as well as potentially financial information if you make purchases on the platform.

Profile information

- **Basic details**: Name, age, birthday, gender, location, and relationship status.
- **Contact information**: Email address, phone number, and your contacts from your address book.
- **Professional details**: Employment history and job information.

Content and activity

- **User-generated content**: Posts, photos, videos, and messages you create and share.
- **Interactions**: Likes, shares, comments, and mentions of other users, pages, or hashtags.
- **Engagement metrics**: How long you watch videos, how often you interact, and the frequency of your use.

Device and technical data

- **Device information**: Operating system, browser type, and unique device IDs.
- **Connection data**: IP address, network, and connection speed.
- **Geolocational data**: Your real-time location, where you live, and places you visit.

16. Discuss the advantages and disadvantages of symmetric key cryptography compared to asymmetric key cryptography? **U/CO3**

Symmetric key cryptography

Advantage	Disadvantage
Speed and efficiency: It is much faster than asymmetric methods because its algorithms are less computationally intensive. This makes it ideal for encrypting large amounts of data like files, databases, and VPNs.	**Key distribution challenge:** The secret key must be securely shared between the sender and receiver before communication can begin. This can be difficult, especially with many participants, and is a major security vulnerability.
Simplicity: It is simpler to implement and requires less computational power.	**Scalability issues:** For every new communication pair, a unique key is needed. This creates a scalability problem, as the number of keys required increases dramatically with the number of users.
Lower overhead: The smaller ciphertext size and faster processing make it more resource-efficient.	**Lack of non-repudiation:** Since both parties have the same key, it's difficult to prove who actually sent a message.

Asymmetric key cryptography

Advantage	Disadvantage
Simplified key distribution: A public key can be shared openly with anyone, while the private key is kept secret, eliminating the need for a pre-shared secret.	**Slower performance:** It is significantly slower than symmetric encryption due to its more complex algorithms.
Enhanced security for key exchange: It provides a secure way to exchange a symmetric key, which can then be used for faster bulk data encryption in a hybrid system.	**Higher computational overhead:** The computational resources required are much greater than for symmetric methods.
Digital signatures and non-repudiation: It enables digital signatures and it provides non-repudiation.	**Larger key sizes:** The keys used in asymmetric cryptography are much larger compared to symmetric keys.

17. How does the principle of integrity apply to data on the internet? **U/CO3**

The principle of integrity on the internet means ensuring data is accurate, consistent, and trustworthy throughout its lifecycle by protecting it from unauthorized modification, corruption, or loss. This is achieved through various technical and procedural safeguards, including encryption, access controls, data validation, checksums, digital signatures, and

regular backups. Maintaining data integrity is crucial for secure online transactions, reliable decision-making, and regulatory compliance.

18. How do encryption and decryption work together to protect sensitive information? **U/CO3**

Encryption and decryption work together by scrambling readable information into an unreadable format using an algorithm and a unique key during encryption, and then using the corresponding key to reverse the process back into its original, readable form during decryption.This two-step process ensures that only authorized parties with the correct key can access sensitive data, protecting it from interception and unauthorized access.

Encryption and Decryption work together to protect information

- **Confidentiality**: Encryption ensures that even if data is intercepted, it remains private because only the correct key can decrypt it.
- **Authentication and integrity**: In some systems, like public-key cryptography, the combination of a public key for encryption and a private key for decryption helps verify that the message came from the intended sender and hasn't been altered.
- **Layered security**: By encrypting data both "at rest" (when stored) and "in transit" (when being sent), organizations add multiple layers of protection against theft and interception.

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